Worksheet: Other Elasticities

ECON 201

Definitions

- **Income elasticity of demand**: the percentage change in demand that would occur in response to a 1% increase in income,

$$\varepsilon_{\text{income}} = \frac{\%\text{ change in demand}}{\%\text{ change in income}}$$

  - **Normal good**: $\varepsilon_{\text{income}} > 0$. People buy more of it as their income rises. There are two types:
    - **Necessity**: $0 < \varepsilon_{\text{income}} < 1$. Demand grows more slowly than income.
    - **Luxury**: $\varepsilon_{\text{income}} > 1$. Demand grows faster than income.
  - **Inferior good**: $\varepsilon_{\text{income}} < 0$. People buy less of it as their income rises.
- **Cross-price elasticity of demand** for good A: the percentage change in demand for A that would occur in response to a 1% increase in the price of good B,

$$\varepsilon_{\text{cross}} = \frac{\%\text{ change in demand for A}}{\%\text{ change in the price of B}}$$

  - **Substitutes**: $\varepsilon_{\text{cross}} > 0$. A rise in the price of B raises demand for A.
  - **Complements**: $\varepsilon_{\text{cross}} < 0$. A rise in the price of B lowers demand for A.

1. Income elasticity

Maya works part time, and her monthly income rises from \$1,000 to \$1,200. Her purchases for the month change as below.

	At \$1,000	At \$1,200	% change in demand	Income elasticity	Normal or inferior?
Packs of ramen	40	32			
Gallons of gas	20	22			
Steak dinners	2	3			

Which of the normal goods is a necessity, and which is a luxury?

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2. Cross-price elasticity

The campus cafe raises the price of a coffee from \$2.50 to \$3.00, and Maya's purchases for the month change as below.

	At \$2.50	At \$3.00	% change in demand	Cross-price elasticity	Substitutes or complements?
Coffee across the street	6	9			
Donuts at the campus cafe	8	6			
Packs of ramen	32	32			

What does ramen's cross-price elasticity tell you?
